UNIT: 4 Virtualization

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4.1.1 Introduction of Virtualization

- ✓ User share the data present in the clouds like application etc, but actually with the help of **virtualization users shares the Infrastructure.**
- ✓ **Virtualization is a technique to divide the computer resources logically.**
- ✓ **Virtualization is technology that you can use to create virtual representations of servers, storage, networks, and other physical machines.**
- ✓ **With the help of Virtualization, multiple operating systems and applications can run on the same machine and its same hardware at the same time, increasing the utilization and flexibility of hardware.**
- ✓ Using cloud, it is very costly to provide physical services per customer on rent because it becomes very costly and user will not use full services. For solution of this, concept of virtualization becomes popular.

4.1.2 Working of Virtualization

- ✓ **Hypervisor software facilitates virtualization**.
- ✓ A hypervisor sits on top of an operating system.
- ✓ Hypervisors take physical resources and divide them up so that virtual environments can use them.
- ✓ A virtual machine created by a hypervisor functions as a single data file, and we can move it from one computer to another, open it there, and it works the same as on any other machine.
  - **Host Machine: The machine on which the virtual machine is going to be built is known as Host Machine.**
  - **Guest Machine: The virtual machine is referred to as a Guest Machine.**

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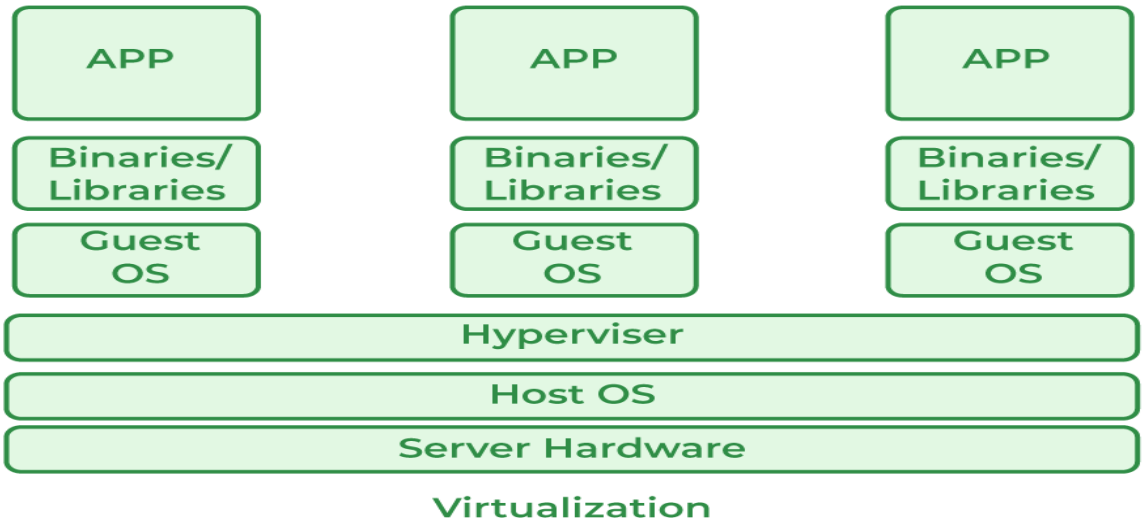

4.1.3 Types of Virtualization

1. Desktop Virtualization
2. Network Virtualization
3. Storage Virtualization
4. Data Virtualization
5. Hardware Virtualization
6. Software Virtualization

Desktop Virtualization:

- ✓ It involves creating virtual instances of a desktop operating system (OS) on a remote server or in a data center.
- ✓ Instead of running the OS and applications directly on a user's local device (such as a desktop or laptop), the processing occurs on a centralized server.
- ✓ Users access their virtual desktops remotely, typically through web browsers, or other endpoint devices.
- ✓ The user device serves as the interface for the end user to interact with the virtual desktop environment.
- ✓ The client software or protocol manages the communication, sending user input to the server and displaying the desktop output on the user's device.
- ✓ **Architecture commonly used in Virtual Desktop Infrastructure (VDI):**
  - **User Device**: The user device is the endpoint device used by the end user to access the Virtualized desktop. This can be a traditional desktop computer, laptop, thin client, or even a mobile device
  - **Client Software/Remote Desktop Protocol (RDP):** Some virtualization solutions require client software installed on the user devices. In other cases, protocols like Remote Desktop Protocol (RDP) or Independent Computing Architecture (ICA) are used to facilitate the communication between the user device and the virtual desktop.

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- **Connection Broker:** It manages and brokers the connections between the user devices and the virtual desktop instances.
- **Virtualization Host Servers:** Virtualization host servers, often referred to as hypervisors, host multiple virtual machines (VMs) or virtual desktops. The hypervisor manages the allocation of resources and ensures the isolation of each virtual desktop instance
- **Virtual Desktops (VMs):** Virtual desktops are instances of desktop operating systems running on the virtualization host servers. Each user typically has their own VM.
- **Storage Infrastructure:** The storage infrastructure stores the virtual desktop images, user data, and application data

Network Virtualization

- ✓ Network Virtualization is a process of logically grouping physical networks and making them operate as single or multiple independent networks called Virtual Networks.
- ✓ Network virtualization is a technology that enables the creation of multiple virtualized network environments on a single physical network infrastructure.
- ✓ This technology provides several benefits, including improved resource utilization, flexibility, and easier network management.

Architecture commonly used in Network Virtualization

- **Virtual Network Layer:** The virtual network layer is created on top of the physical network, infrastructure. It allows the creation of multiple independent virtual networks, each with its own set of policies, configurations, and addressing schemes.
- **Hypervisor or Network Virtualization Platform:** A hypervisor or network virtualization platform is responsible for managing the virtual networks. It abstracts the physical network hardware and controls the creation, allocation, and management of virtual network resources.
- **Virtual Switching:** Virtual switches operate at the software level and connect virtual machines (VMS) within the same virtual network. These switches direct network traffic between virtual machines and physical networks.
- **Isolation and Segmentation:** Isolation ensures that each virtual network operates independently. Virtual networks are segmented from each other, preventing interference and providing security and privacy for the data and communication within each virtual network
- **Overlay Networks:** Overlay networks create logical connections over the physical network, allowing communication between virtual machines regardless of their physical location.
- **Dynamic Resource Allocation:** Network virtualization enables dynamic allocation of network resources based on demand. Resources such as bandwidth and routing configurations can be adjusted dynamically to accommodate changing network requirements
- **Network Services and Policies:** Network services, such as firewalls, load balancers, and intrusion detection systems, can be implemented at the virtual network level.

Storage Virtualization

- ✓ It is a technology that abstracts physical storage resources and presents them as a single, unified virtual storage pool.
- ✓ Storage virtualization provides a centralized management interface for administrators to configure, allocate, and monitor storage resources.

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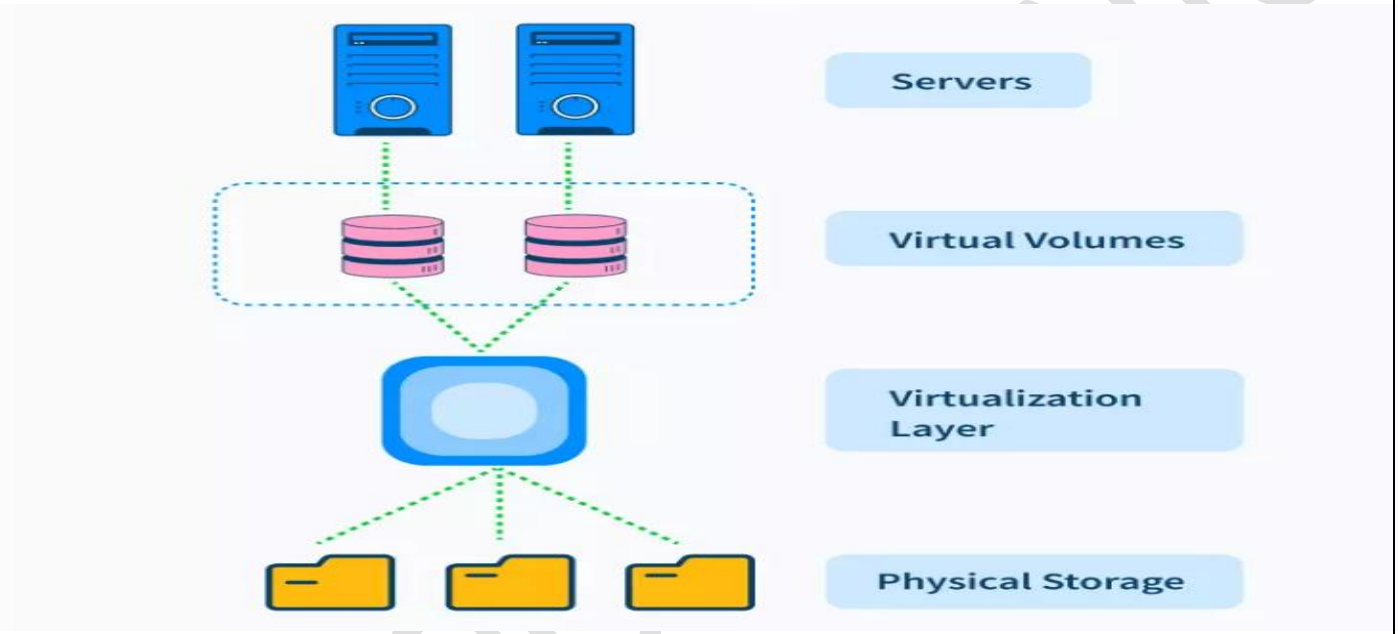
- ✓ Storage virtualization helps improve efficiency, scalability, and flexibility in storage infrastructure.
- ✓ Storage virtualization is implemented through specialized software.
- ✓ This software manages the mapping of logical storage to physical storage, handles storage requests from applications or servers, and provides a unified view of the storage infrastructure.
- ✓ It creates a pool of storage capacity that can be dynamically allocated and managed based on the needs of applications and users.
- ✓ Storage virtualization enables the movement of data between different storage devices without disruption.
- ✓ Storage virtualization can enhance performance by providing features like caching and optimization of data placement



Key Benefits of Storage Virtualization

- ✓ Presents better space utilization and storage options for unorganized data and resources.
- ✓ Thin provisioning is a feature of storage virtualization that allows administrators to allocate storage space on-demand rather than pre-allocating it.
- ✓ Enhances the data backup by enabling snapshots of storage resources.
- ✓ Provides an easy copy of data from one appliance to another at any specified location.
- ✓ With a centralized system to manage numerous clients and vendors, storage virtualization helps preserve time as well as money.
- ✓ Allows migration of data to be completed quickly with fewer or no barriers to the movement.
- ✓ Reduces the need for hardware for storage, making it an easier and cheaper option for storage.

Major Challenges of Storage Virtualizations

- ✓ Storage systems need frequent up gradation to meet the challenging nature of applications and huge data.
- ✓ Managing the different software and hardware can get difficult when there are several hardware and software elements.
- ✓ Accessing data with storage virtualization, there is always a risk of cyber-attacks and various cyber threats in virtual environments.

- ✓ As many a time, it happens when vendors provide storage solutions but ignore the complexities of backing up virtual storage pools.
- ✓ Similarly, they fall in situations when there is a need for immediate recovery of data in case of hardware failure or any other issue.
- ✓ Storage virtualization can lead to access issues. This can be if the LAN connection is disrupted, or internet access is lost due to some reason.
- ✓ When there is a need to switch from a smaller network to a larger one because of the capacity of the current one. The migration process is time-consuming and can even result in downtime.

Data Virtualization

- ✓ Data virtualization is a technology that allows an organization to access and manipulate data without necessarily having to physically move or replicate it.
- ✓ It provides a layer of abstraction over disparate data sources, such as databases, applications, and cloud services, allowing users to view and interact with the data as if it were a single, unified source.
- ✓ Data virtualization creates a virtual representation of data from different sources.
- ✓ Data virtualization integrates data from different types of databases, applications, and other sources, making it appear as if it comes from a single source.
- ✓ It enables real-time access to data, ensuring that the information is up-to- date and accurate
- ✓ It allows organizations to adapt quickly to changes in data sources and requirements without significant disruptions.

Key points about data virtualization:

- ✓ **Integration of Diverse Data Sources:** Data virtualization integrates data from different types of databases, applications, and other sources, making it appear as if it comes from a single source.
- ✓ **Real-time Access:** It enables real-time access to data, ensuring that the information is up-to-date and accurate
- ✓ **Reduced Data Redundancy**: Unlike traditional methods that involve duplicating data, data virtualization minimizes the need for data replication, reducing storage costs and data inconsistencies.
- ✓ **Agile and Flexible:** Allows organizations to adapt quickly to changes in data sources and requirements without significant disruptions.

Layers of Data Virtualization

Connection layer: With the use of connectors and communication protocols, this layer is in charge of accessing the data across numerous source systems that contain both organized and unstructured data. Platforms for data virtualization can connect to various data sources, such as SQL and NoSQL databases like MySQL, Oracle and MongoDB etc.

Abstraction layer: The abstraction layer, also known as the **virtual layer**, serves as
a link between all data sources and all business users, forming the backbone of the entire virtualization system.

Consumption layer: A single point of access to the data stored in the underlying sources is offered by a different tier of the data virtualization architecture.

Advantages of Data Virtualization:

Speed: Quick access to diverse data without moving it around.

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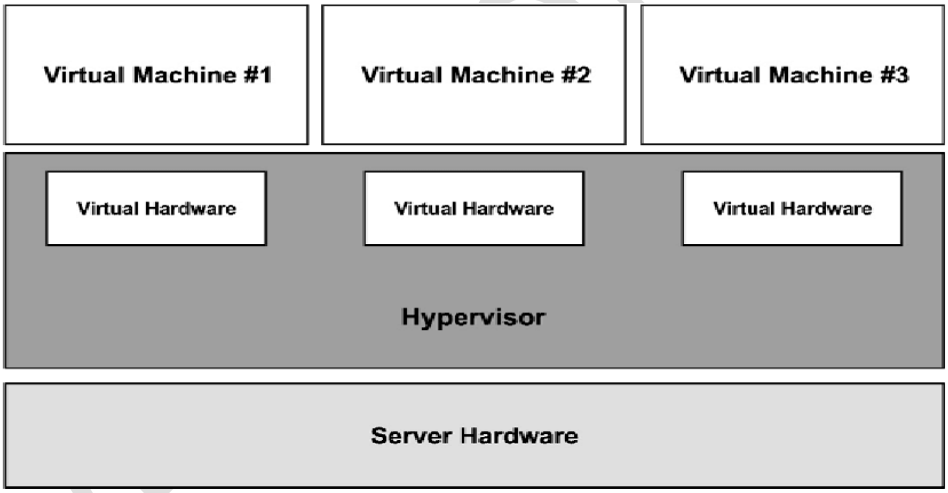
Cost-Efficiency: Reduces duplication and storage costs.
Unified View: Provides a single view of data from various sources.
Real-time Integration Supports real-time data updates.
Agility: Easily adapts to changes in data sources.

Disadvantages of Data Virtualization:
Performance: May face performance issues with large datasets.
Network Dependency: Relies on efficient network performance.
Security Challenges: Handling security and data privacy can be complex
Complexity: Implementing and managing can be intricate
Limited Offline Access: Real-time reliance may limit functionality offline.

Hardware Virtualization

- ✓ Hardware virtualization has a base layer-a hypervisor.
- ✓ This layer is loaded directly on the bare server and acts as an Intermediary between the server hardware and virtual machines.
- ✓ Parallels Cloud Server is based on the concept of hardware virtualization.
- ✓ To allocate hardware and resources to virtual machines, Parallels Cloud Server virtualizes all hardware on the server.
- ✓ Once virtualized, hardware and resources can be easily assigned to virtual machines.
- ✓ Based on the virtual hardware, a virtual machine runs its own complete copies of an operating system and applications.



Benefits of Hardware Virtualization:

- Create multiple virtual machines with different operating systems on a single physical computer
- Run several guest operating systems and their applications simultaneously on a single physical computer without rebooting
- Virtualized the computing environment, reduce hardware costs, lower operating expenses, and increase productivity.

Key aspects of hardware virtualization:

Hypervisor (Virtual Machine Monitor): A fundamental component of hardware virtualization is the hypervisor, or virtual machine manager (VMM). It is a software layer that enables the creation and management of virtual machines (VMs) on a physical computer or server.

Virtual Machines (VMs): Virtualization creates virtual machines, which are independent instances of operating systems running on the same physical hardware. Each VM operates as if it were running on a dedicated physical machine, with its own OS, applications, and resources.

Resource Allocation: The hypervisor allocates physical resources such as CPU, memory, and storage to each virtual machine, allowing multiple VMs to coexist on a single physical server

Isolation: Virtualization provides isolation between virtual machines. This means that activities or issues in one VM do not affect others, enhancing security and stability.

Snapshot and Migration Virtualization: It allows the creation of snapshots, which are pointing time images of a virtual machine. This feature is useful for backup, recovery, and testing

Server Consolidation: Hardware virtualization facilitates server consolidation by running multiple virtual machines on a single physical server. It optimizes resource utilization and reduces the number of physical servers needed.

Testing and Development: Virtualization is widely used in testing and development environments, allowing developers to create and test software in isolated VMs without affecting the production environment.

Advantages of Hardware Virtualization:

- **Resource Efficiency:** Optimizes hardware usage by running multiple virtual machines on one server.
- **Isolation:** Ensures each virtual machine operates independently for enhanced security and stability
- **Flexibility:** Allows different operating systems and applications to coexist on the same hardware
- **Dynamic Resource Allocation:** Efficiently assigns and adjusts resources based on workload demands.
- **Snapshot and Migration:** Supports backups, disaster recovery, and seamless VM movement between servers.

Disadvantages of Hardware Virtualization:

- **Overhead**: Introduces some performance overhead due to the hypervisor layer.
- **Complexity**: Requires skilled management, configuration errors can impact performance
- **Compatibility Issues**: Not all hardware may be compatible with virtualization technologies.
- **Single Point of Failure**: Hypervisor failure can affect all virtual machines on a host.
- **Performance Limitations**: Certain high-performance workloads may experience reduced efficiency.
- **Security Concerns**: Vulnerabilities in the hypervisor create security risks.

Software Virtualization

✓ Example of software virtualization is VMware software, virtual box etc.

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- ✓ **Software virtualization is of three types**:

Operating System Virtualization:

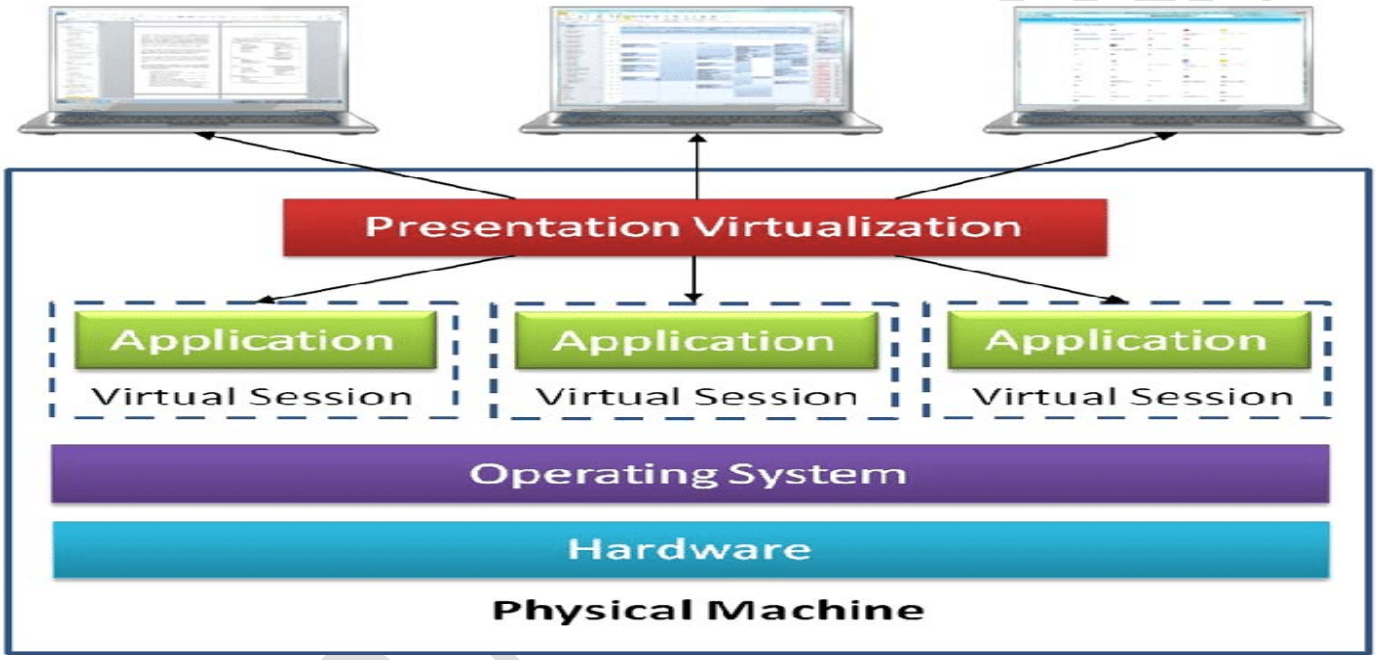
- Here, All operating systems work independently to accomplish the task. A particular operating system performs its specified job.
- Example- VMware software.

Application Virtualization:

- Here, it encapsulates different parallel running programs within an operating system, and users can remotely access their applications from the main server.
- Example- VMware horizon

Service Virtualization:

- This is a type of virtualization that simulates the behavior of some component in the combination component-based application.



key points about software virtualization:

- Application Virtualization: Software virtualization often involves isolating and running applications in a virtualized environment.
- Operating System Virtualization: In some cases, software virtualization can also be used to create virtualized instances of entire operating systems.
- Compatibility: Software virtualization can be employed to ensure compatibility of applications with different operating systems. For example, running a Windows application on a Linux system through virtualization.
- Resource Partitioning: It allows for the partitioning of resources within an operating system to create isolated environments for running applications.
- Portability: Virtualized software environments can be more portable and easily moved between different systems, as they are not as tightly bound to the specifics of the underlying hardware on operating system.
- Efficiency: Application virtualization can improve resource utilization by allowing multiple applications to run on the same system without conflicts

Advantages of Software Virtualization:

- Compatibility: Runs applications on different operating systems.
- Efficiency: Optimizes resource use within an operating system
- Security: Provides isolated environments, enhancing system security.
- Portability: Easily moves virtualized environments between systems.

Disadvantages of Software Virtualization:

- Performance Overhead: May introduce some performance impact
- Dependency Management: Handling dependencies can be complex.
- Not for All Workloads: Some resource-intensive tasks may not perform optimally.
- Compatibility Issues: Some applications may not be suitable for virtualization.

4.1.4 Advantages and Disadvantage of Virtualization

Advantages of Virtualization:

Resource Optimization: Maximizes hardware utilization by running multiple virtual instances on a single physical server. It leads to cost savings in terms of hardware, power, cooling, and space

High Availability: One of the main benefits of Virtualization is that it provides advance features which allow virtual instances to be available all the times.

Disaster Recovery is efficient and easy: With the help of virtualization Data Recovery, Backup, Duplication becomes very easy. With the tools of Virtualization real time data backup recovery and mirroring become easy task and provide zero percent data loss.

Quick and Easy Set up: In traditional methods Setting up physical system and servers are very time-consuming. Firstly Purchase them in bulk after that wait for shipment. When Shipment is done then wait for Setting up and after that again spend time in installing required software etc. Which will consume very time But with the help of virtualization the entire process is done in very less time which results in productive setup.

Cost Savings: Reduces hardware and energy costs by consolidating multiple virtual environments on fewer physical servers.

Testing and Development: Ideal for creating isolated environments for testing and development purposes.

Disadvantage of Virtualization

High Initial Investment: The initial setup costs for storage and servers can be higher than a traditional setup.

Complexity: Managing virtualized environments can be complex, especially as the number of VMs increases.

Security Risks: Virtualization introduces additional layers, which may pose security risks if not properly configured and monitored.

Learning New Infrastructure: As Organization shifted from Servers to Cloud. They required skilled staff who can work with cloud easily. Either they hire new IT staff with relevant skill or provide training on that skill which increases the cost of company.

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Data can be at Risk: Working on virtual instances on shared resources means that our data is hosted on third party resource which put's our data in vulnerable condition. Any hacker can attack on our data or try to perform unauthorized access. Without Security solution our data is in threaten situation**.**

Single Point of Failure: If the hypervisor fails, it can impact all VMs running on the host.

4.1.5 Characteristic and Applications of Virtualization

Characteristic of Virtualization

Sharing: Virtualization allows the creation of a separate computing environment within the same host. This basic feature is used to reduce the number of active servers and limit power consumption.

Resource Pooling: Physical resources such as CPU, memory, and storage are pooled and dynamically allocated to virtual instances based on demand.

Encapsulation: Applications, operating systems, or entire environments are encapsulated into virtual containers or machines, making them portable and independent.

Compatibility: Virtualization allows running multiple operating systems and applications on the same physical hardware improving cross-platform compatibility.

Flexibility: Virtualization provides flexibility by allowing easy scaling, migration, and adaptation to changing workloads or requirements.

Snapshot and Cloning: Virtualization enables the creation of snapshots and clones of virtual instances for backup, testing. and rapid deployment

Dynamic Resource Allocation: Resources can be dynamically adjusted to meet changing demands, ensuring efficient utilization

Hypervisor: The hypervisor, or Virtual Machine Monitor (VMM), is a key component that manages virtual machines, allocating resources and ensuring isolation

Live Migration: Virtual machines can be moved between physical servers without downtime, supporting load balancing and maintenance without service interruption

High Availability: Virtualization supports features like failover and high availability to ensure continuous service availability in case of hardware failures.

Resource Efficiency: Virtualization optimizes hardware usage by allowing multiple virtual instances to share the same physical resources

Portability: Virtualized environments are often more portable, allowing easy movement between different systems or cloud platforms.

Applications of Virtualization

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Server Virtualization: Run multiple virtual servers on a single physical server to optimize resource utilization. Ensure continuous server operation through features like live migration and failover

Desktop Virtualization: Manage and deploy desktop environments from a central server. Provide isolated virtual desktops for users, enhancing security and customization

Application Virtualization: Run applications on different operating systems without conflicts. It reduces dependencies.

Network Virtualization: Create isolated virtual networks to enhance security and segmentation. It enables virtual network functions (VNFs) for dynamic network configuration.

Disaster Recovery: Use virtualization for quick and efficient backups and disaster recovery.Test disaster recovery plans in virtualized environments without affecting the production system.

Education and Training: Provide virtual labs for educational purposes without the need for physical hardware.

Testing and Development: Create isolated environments for testing software and applications. It use snapshots and clones for efficient testing and development processes.

Difference between Cloud Computing and Virtualization

Cloud Computing	Virtualization
Cloud computing is used to provide pools and automated resources that can be accessed on-demand	It is used to make various simulated environments through a physical hardware system
Cloud computing setup is tedious, complicated.	Virtualization setup is simple as compared to cloud computing.
Cloud computing is high scalable.	Virtualization is low scalable compared to cloud computing
Cloud computing is Very flexible.	Virtualization is less flexible than cloud computing
In the condition of disaster recovery, cloud computing relies on multiple machines.	Virtualization it relies on a single peripheral device
The workload is stateless	The workload is Stateful
The total cost of cloud computing is higher than Virtualization	The total cost of virtualization is lower than Cloud Computing
Cloud computing is of two types: Public cloud and Private cloud.	Virtualization is of two types Hardware virtualization and Application virtualization
Cloud computing provides unlimited storage space.	In Virtualization storage space depends on physical server capacity in virtualization

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4.2.1 Technologies of virtualization: Hypervisors

hypervisor:

- ✓ A hypervisor is also known as Virtual Machine Monitor (VMM).
- ✓ It is a software layer that enables the creation and management of virtual machines (VMs) on a physical computer or server.
- ✓ The primary purpose of a hypervisor is to allow multiple operating systems (OS) or applications to run on a single physical machine concurrently, with each operating system and its associated applications functioning as if they were running on a dedicated, independent server.

How Does a Hypervisor Work?

- ✓ They operate on the server level and can be either physical or virtual, and hardware or software-defined
- ✓ They load the client operating systems of the VMs
- ✓ It distributes the computing resources such as CPU, memory, bandwidth, and disk storage for each of the VMs. It does this by creating pools of hardware resources, which it then allocates to VMs
- ✓ VMs can create requests for the hypervisor through API calls

Key aspects of Hypervisor:

- ✓ **Virtual Machines (VMs):** Virtualization creates virtual machines, which are independent instances of operating systems running on the same physical hardware. Each VM operates as if it were running on a dedicated physical machine, with its own OS, applications, and resources.

- ✓ **Resource Allocation**: The hypervisor allocates physical resources such as CPU, memory, and storage to each virtual machine, allowing multiple VMs to coexist on a single physical server

- ✓ **Isolation:** Virtualization provides isolation between virtual machines. This means that activities or issues in one VM do not affect others, enhancing security and stability.

- ✓ **Snapshot and Migration Virtualization**: It allows the creation of snapshots, which are pointing time images of a virtual machine. This feature is useful for backup, recovery, and testing

Basically, there are two types of Hypervisors: TYPE-1Hypervisor: and Type 2 Hypervisor

TYPE-1Hypervisor:

- ✓ It is also known as a “Native Hypervisor” or “Bare metal hypervisor”.
- ✓ The hypervisor runs directly on the underlying host system.
- ✓ Installs directly on the hardware, managing virtual machines without the need for a host operating system.
- ✓ It does not require any base server operating system.
- ✓ It has direct access to hardware resources.
- ✓ Type I is the best for enterprise computing and large-scale deployments.

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- ✓ Examples: vSphere,XenServer, Hyper-V

The advantages of Type I hypervisors are:

Scalability: The ability to directly assign resources makes Type I scalable.

Optimization of physical resources: Type I allows optimization of single server hardware, thus freeing up data costs and cutting down energy usage.

Better Resource Allocation: Type I hypervisors give administrators the ability to set resource allocation manually.

Type 2 Hypervisor

- ✓ It is also known as Hosted hypervisor or embedded Hypervisor.
- ✓ It runs on top of the host OS.
- ✓ It supports Operating System virtualization.
- ✓ It provide less hardware resource utilization
- ✓ Incur overhead as hypervisor runs on top of the supported OS.
- ✓ Less secure due to a software based hypervisor
- ✓ Examples: VM ware Workstation, VMware Player, Oracle Virtual box.
- ✓ Type II hypervisors also support multiple guest machines but are not allowed to directly access the host hardware and its resources.

The advantages of Type 2 hypervisors are:

Easier setup: These hypervisors are easier to set up and manage

Easier management: Type 2 hypervisors do not require a dedicated administrator

Compatibility: Type 2 hypervisors are compatible with a wider range of hardware, since they run on top of an OS, instead of specific hardware machines.

Choosing the right hypervisor

- ✓ Type 1 hypervisors offer much better performance than Type 2
- ✓ One of the best ways to find out which hypervisor meets your needs is to compare their performance metrics.
- ✓ The following factors must be examined before selecting the appropriate hypervisor.

Understand your needs: The data center (and your job) for the company and its applications. In addition to the requirements of your company, you (and your IT staff) also have your own requirements.

- Flexibility
- Scalability
- Usability
- Availability
- Reliability
- Efficiency
- Reliable support

The cost of a hypervisor: For many buyers, a hypervisor is the most difficult thing in selecting the right balance between cost and functionality.

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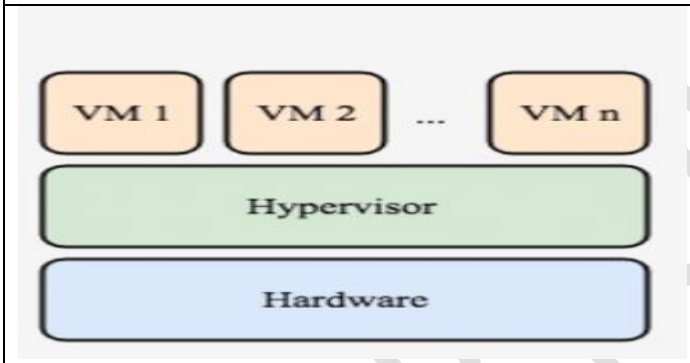
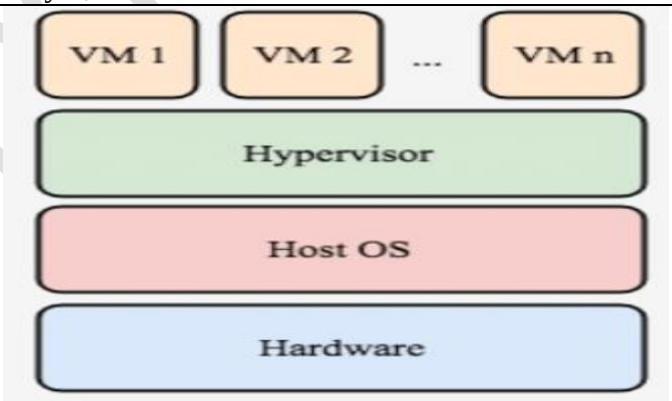
Virtual machine performance: Virtual systems should achieve or exceed, in relation at least, to the server applications, their physical counterpart's performance

Ecosystem: The role that an ecosystem's hypervisor can have in deciding on whether a solution is cost-effective or not is tempting to ignore that is, availability of documentation, support, training, development and consultancy services

Test for yourself: You can obtain basic experience from your existing desktop or laptop. To build a nice virtual learning and testing environment, you can run VMware vSphere and Microsoft Hyper-V in either VMware Workstation or VMware Fusion.

Differentiate between Type-1 and Type-2 Hypervisor.

Type-1 Hypervisor	**Type-2 Hypervisor**
Also known as Bare metal hypervisor.	Also known as Hosted hypervisor.
Directly runs on server Hardware	Runs on top of the supported OS
Support Hardware virtualization	Support Operating System virtualization
Hard to Set up	Easy to Set up
Provide better hardware resource utilization	Provide less hardware resource utilization
Minimizing overhead due to direct hypervisor interaction with hardware resources.	Incur overhead as hypervisor runs on top of the supported OS.
More secure due to hardware based hypervisor	Less secure due to a software based hypervisor
Examples: vSphere,XenServer, Hyper-V	Examples: VM ware Workstation, VMware Player, Oracle Virtual box.